Raj Sanjay Shah

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About me

PhD student in Interactive Computing at Georgia Tech, working on training and adapting large language models for cognitively grounded and high-stakes domains. My work spans pretraining from scratch (e.g., FLANG; BabyLM), large-scale post-training (domain adaptation, steerability for mental health systems), and dynamic unlearning frameworks for model control. I study how training objectives, data distributions, and architectural constraints shape reasoning, development, and generalization in LLMs. My research combines large-scale model training, evaluation, and theory-driven analysis, with publications in ACL, NAACL, EMNLP, COLM, CHI, and CSCW.
More recently, I have been building a foundation model for cognitive science, designing a structured training stack that integrates general pre-, mid-, and post-training with human plausibility fine-tuning and contrastive multi-objective alignment to transform large language models into experimentally tractable systems for probing the structure and development of human-like cognition.

Education

Aug 2023 – | **Georgia Institute of Technology**
*Ph.D in **Computer Science*** (GPA: 4.0/ 4.0)
Advisor: Professor Sashank Varma

May 2024 – Sep 2025 | **Stanford University**
Visiting Student Researcher
Advisor: Professor Diyi Yang

Aug 2021 – May 2023 | **Georgia Institute of Technology**
*MS in **Computer Science** (GPA: 4.0/ 4.0, Specialization: Machine Learning)*

Aug 2017 – Jul 2021 | **Birla Institute of Technology and Science, Pilani, India**
Bachelor of Engineering in Computer Science (CGPA: 8.54 / 10)
Minor in Data Science (Minor GPA: 9.8 / 10)

Employment

May 2025 – Aug 2025 | **Amazon Science** *(Applied Scientist Intern II)*
Developed a clarification-driven dialogue summarization framework that identifies information gaps and generates targeted questions to improve summary completeness and accuracy across clinical, legal, and customer support conversations.

May 2024 – Aug 2024 | **Amazon Science - Healthscribe Team** *(Applied Scientist Intern II)*
Assessing Large Language Models for high-quality behavioral therapy progress note generation. Deployed in Amazon AWS.

May 2023 – Aug 2023 | **IBM Research - Human-Centered AI Team** *(Sr. Research Software Engineer - AI)* Explored different intervention strategies for high-quality human-AI collaborative data creation. Built a full-stack application for user studies. User studies investigate the idea of appropriate user reliance on AI agents using subtle real-time nudges.

May 2022 – Aug 2022 | **Epic Systems -** *(Software Developer - ML Intern)*
Trained Large Language Models to predict workflow actions of Health Care Providers with the goal of bringing relevant patient information to the Provider Dashboard on the fly.

Jan 2021 – Jun 2021 | **JPMorgan Chase -** *(Software Development Intern)* Worked in the Merchant Services team on developing automated testing scripts for the onboarding platform using the Postman tool.

May 2020 – Jun 2020 | **Samsung R&D Institute, India -** *(Research Intern)*
Developed a bidirectional Long-short-term memory-based variable autoencoder to generate melodies and chords for piano music as a part of the User-Experience: Ringtone project.

Honors & Awards

IndoML Early Career Award *2025*
Recognized as an outstanding early-career researcher (final-year PhD / recent graduate) in a competitive selection process.

Georgia Tech President's Fellowship *2023-2026*
This award is given to students who have excelled in scholarship and are pursuing doctoral degrees. Selection for this competitive award is based on multiple areas, including but not limited to academic history, standardized test scores, and letters of recommendation. This recognition comes with a stipend of $5,500 per academic year, for a total of 3 years.

Georgia Tech Students Honors Program - The Marshall D. Williamson Fellowship Award *2023*
Awarded to a well-rounded, second-year Master's student who best embodies Marshall's values of academic excellence and leadership.

Georgia Tech Students Honors Program - Donald V. Jackson Fellowship Award *2022*
Awarded to a well-rounded, first-year Master's student who best embodies the values of academic excellence and leadership.

Lalita and Ravindra Nath (LRN) Foundation for Student Travel Grant *2022*
Awarded for travel to EMNLP 2022. The award is given to selected student candidates with an accepted paper as a first author at a top-tier International conference outside India and a paper or poster in the COMSNETS conference or one of the associated workshops (in any of the previous COMSNETS).

Birla Institute of Technology and Science best paper award - information retrieval *2020*
Awarded to a computer science undergraduate for rigor in research and academics.

Publications

2026

1. [Patent] Brachman, M. *et al. Guiding a user to interact with an intelligent computing system using best practices* 2026.
2. Arnaout, H. *et al.* Responsible Evaluation of AI for Mental Health. *arXiv preprint arXiv:2602.00065* (2026).
3. Choshen, L. *et al.* BabyLM Workshop: Call for Papers for the 2026 BabyLM Workshop. *arXiv preprint arXiv:2602.20092* (2026).
4. **Shah, R. S.**, Lalai, H. N., Pfister, H., Varma, S. & Guo, G. *When Visuals Aren't the Problem: Evaluating Vision-Language Models on Misleading Data Visualizations* in *Under Review at ARR* (2026).
5. **Shah, R. S.**, Warstadt, A., Frank, M. & Varma, S. On the use of pre-trained language models in cognitive science. *arXiv preprint arXiv:2501.12651* (2026).
6. **Shah, R. S.** *et al.* Simulating AI Patients for Psychotherapy: Challenges and Opportunities (2026).
7. 🏆 Honorable Mention: Louie, R. *et al.* Can LLM-Simulated practice and feedback upskill human counselors? a randomized study with 90+ novice counselors. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (2026).

2025

8. Charpentier, L. *et al.* BabyLM Turns 3: Call for papers for the 2025 BabyLM workshop. *arXiv preprint arXiv:2502.10645* (2025).
9. Gupta, S., **Shah, R. S.** & Varma, S. *Do LLMs Suppress Naïve Theories? Investigating Scientific Reasoning and Development in GPT-4o* in *Advances in Cognitive Systems* (2025).
10. Hsu, S.-L. *et al.* Helping the Helper: Supporting peer counselors via AI-Empowered practice and feedback. *Proceedings of the ACM on Human-Computer Interaction* **9,** 1–45 (2025).
11. Jun, J., Marupudi, V., **Shah, R. S.** & Varma, S. *A Neural Network Model of Complementary Learning Systems: Pattern Separation and Completion for Continual Learning* in *Proceedings of the Annual Meeting of the Cognitive Science Society* **47** (2025).
12. Lalai, H. N., Ramakrishnan, A. A., **Shah, R. S.** & Lee, D. *From Intentions to Techniques: A Comprehensive Taxonomy and Challenges in Text Watermarking for Large Language Models* in *Findings of the Association for Computational Linguistics: NAACL 2025* (2025), 6147–6160.
13. **Shah, R. S.**, Huang, J., Murugesan, K., Baracaldo, N. & Yang, D. The Unlearning Mirage: A Dynamic Framework for Evaluating LLM Unlearning. *Second Conference on Language Modeling* (2025).
14. **Shah, R. S.** *et al.* The World According to LLMs: How Geographic Origin Influences LLMs' Entity Deduction Capabilities. *Second Conference on Language Modeling* (2025).
15. **Shah, R. S.** *et al.* TN-Eval: Rubric and Evaluation Protocols for Measuring the Quality of Behavioral Therapy Notes. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)* (2025).

2024

16. Chaszczewicz, A. *et al. Multi-Level Feedback Generation with Large Language Models for Empowering Novice Peer Counselors* in *Association for Computational Linguistics* (2024).
17. Du, J. *et al. LLMs Assist NLP Researchers: Critique Paper (Meta-) Reviewing* in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (2024), 5081–5099.
18. Li, A. *et al. Incremental Comprehension of Garden-Path Sentences by Large Language Models: Semantic Interpretation, Syntactic Re-Analysis, and Attention* in *Proceedings of the Annual Meeting of the Cognitive Science Society* **46** (2024).
19. **Shah, R. S.**, Bhardwaj, K. & Varma, S. *Development of Cognitive Intelligence in Pre-trained Language Models* in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (2024), 9632–9657.
20. **Shah, R. S.**, Gandhi, A., Marupudi, V. & Varma, S. *Mitigating Catastrophic Interference using Power Law Training Environments* in *27TH EUROPEAN CONFERENCE ON ARTIFICIAL INTELLIGENCE* (2024), (Equal contribution between the first two authors).
21. **Shah, R. S.**, Guo, G., Kang, J. J., Pfister, H. & Varma, S. *Understanding Graphical Perception in Data Visualization through Zero-Shot Prompting of Vision-Language Models* in *NeurIPS 2024 Workshop on Behavioral Machine Learning* (2024).
22. Vemuri, S., **Shah, R. S.** & Varma, S. *Evaluating Typicality in Combined Language and Vision Model Concept Representations* in *Proceedings of the Annual Meeting of the Cognitive Science Society* (2024).
23. Yang, W. *et al. What Makes Digital Support Effective? How Therapeutic Skills Affect Clinical Well-Being* in *Proceedings of the ACM on Human-Computer Interaction 6* (2024).

2023

24. Bhardwaj, K., **Shah, R. S.** & Varma, S. *Pre-training LLMs using human-like development data corpus* in *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning* (eds Warstadt, A. *et al.*) (Association for Computational Linguistics, Singapore, Dec. 2023), 339–345.
25. **Shah, R. S.**, Marupudi, V., Koenen, R., Bhardwaj, K. & Varma, S. *Numeric Magnitude Comparison Effects in Large Language Models* in *Findings of the Association for Computational Linguistics: ACL 2023* (Association for Computational Linguistics, Toronto, Canada, July 2023), 6147–6161.
26. Wang, T. *et al. Metrics for Peer Counseling: Triangulating Success Outcomes for Online Therapy Platforms* in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Association for Computing Machinery, Hamburg, Germany, 2023).

2022

27. Pathak, A., **Shah, R. S.**, Kumar, V. & Jakhotiya, Y. JARVix at SemEval-2022 Task 2: It Takes One to Know One? Idiomaticity Detection using Zero and One Shot Learning. *CoRR* **abs/2202.02394.** arXiv: 2202.02394 (2022).
28. **Shah, R. S.** *et al. Modeling Motivational Interviewing Strategies On An Online Peer-to-Peer Counseling Platform* in *Proceedings of the ACM on Human-Computer Interaction 6* (2022).
29. **Shah, R. S.** *et al. When FLUE Meets FLANG: Benchmarks and Large Pretrained Language Model for Financial Domain* in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (Association for Computational Linguistics, 2022).

2021

30. **Shah, R. S.**, Bhatia, A., Gandhi, A. & Mathur, S. *Bitcoin Data Analytics: Scalable techniques for transaction clustering and embedding generation* in *2021 International Conference on COMmunication Systems NETworkS (COMSNETS)* (2021), 1–6.

2020

31. Kristiansen, L.-M., Agarwal, V., Franke, K. & **Shah, R. S.** *CTI-Twitter: Gathering Cyber Threat Intelligence from Twitter using Integrated Supervised and Unsupervised Learning* in *2020 IEEE International Conference on Big Data (Big Data)* (2020), 2299–2308.
32. **Shah, R. S.** & Bhatia, A. *Bitcoin Data Analytics: Exploring Research Avenues and Implementing a Hadoop-Based Analytics Framework* in *Web, Artificial Intelligence and Network Applications* (eds Barolli, L., Amato, F., Moscato, F., Enokido, T. & Takizawa, M.) (Springer International Publishing, Cham, 2020), 178–189.

Teaching Experience

Jan 2026 - May 2026
CS 6795: Grad Intro to Cognitive Science, Georgia Tech
Graduate Teaching Assistant - Professor Keith McGreggor

Aug 2025 - Dec 2025
CS 7641: Human and Machine Learning, Georgia Tech
Graduate Teaching Assistant - Professor Sashank Varma
- Created homework and assignments for the course CS 7641.
- Assisted students in their course capstone projects.

Aug 2025 - May 2025
CS 6795: Grad Intro to Cognitive Science, Georgia Tech
Graduate Teaching Assistant - Professor Keith McGreggor

Jan 2024 - May 2024
CS 3790: Intro to Cognitive Science, Georgia Tech
Graduate Teaching Assistant (TA) - Professor Sashank Varma
- Created homework and assignments for the course CS 3790.

Aug 2023 - Dec 2023
CS 7641: Human and Machine Learning, Georgia Tech
Graduate Teaching Assistant - Professor Sashank Varma
- Assisted students in their course capstone projects.

Jan 2023 - May 2023
CS 3790: Intro to Cognitive Science, Georgia Tech
Graduate Teaching Assistant - Professor Sashank Varma
- Created homework and assignments for the course CS 3790.

Aug 2022 - Dec 2022
CS 4650: Natural Language Processing, Georgia Tech
Graduate Teaching Assistant - Professor Alan Ritter
- Making coding exercises for students to have hands-on NLP experience.

Aug 2020 - Dec 2020
CS F469: Information Retrieval, Birla Institute of Technology and Science, Pilani
Undergraduate Teaching Assistant - Professor Vinti Agarwal
- Created a term paper-based assignment to promote student exposure to research.

Service

IndoML Early Career Chair *2026*

Serving as Chair for the IndoML Early Career Program, organizing mentorship, programming, and community-building initiatives for early-career researchers.

Ombudsperson - Interactive Computing, Georgia Institute of Technology *2023 - 2026*
My primary role as the Student Ombudsperson/Neutral Third Party is to serve as a resource for PhD students who have concerns about a faculty or staff member or an issue within the PhD program.

Area Chair for the following conferences
Ethics Area Chair for Industry Track in the Association of Computational Linguistics 2026 and Empirical Methods in Natural Language Processing 2025.

Reviewer for the following conferences
Association of Computational Linguistics Rolling Review 2026, 2025, 2024; CogSci 2026, 2025; International Conference on Machine Learning 2026, Computer-Supported Cooperative Work And Social Computing 2025, 2024, 2023; ACM CHI Conference on Human Factors in Computing Systems 2026, 2025, 2023; Empirical Methods in Natural Language Processing 2022; IEEE BigData 2021.

Student Volunteer for the following conferences
Advances in Cognitive Systems 2025; Computer-Supported Cooperative Work And Social Computing 2022, 2021; Annual Meeting of the Association for Computational Linguistics 2021.

Workshop Organization
BabyLM 2025, 2026 Contributed to the organization of the BabyLM Workshop, supporting community-building efforts around sample-efficient language model research. Responsible for training baselines.
ATL SIGCHI 2023 Contributed to the first edition organization of the ATL SIGCHI, supporting community-building efforts around human-computer interaction.

Mentorship

- Harsh Lalai - Undergraduate @ BITS Pilani (Aug 2023 - Present, Incoming PhD @ Johns Hopkins)
- Jenna Kang - PhD @ NYU (August 2024 - Present)
- Ifdita Hasan Orney - Undergraduate senior @ Stanford (August 2024 - Present)
- Keane Zhang - MS CS @ GT (August 2025 - Jan 2026)
- James Jun - MS CS @ GT (January 2024 - May 2025, Georgia Tech Research Institute)
- Khushi Bhardwaj - BS @ GT (August 2022 - May 2024, Research Scientist at NVIDIA)
- Atith Gandhi - MS CS @ GT (January 2023 - May 2024, SWE at Oracle)
- Ananjan Nandi - MS CS @ Stanford (August 2023 - May 2024, PhD Student at Stanford)
- Siddhant Narang - MS CS @ GT (January 2024 - May 2024, Senior SWE at BlackRock)
- Austin Peng - MS CS @ GT (January 2024 - May 2024, Machine Learning Engineer at Workday)
- Mihir Sharma - BS/MS CS @ GT (January 2024 - May 2024, Research Engineer at NVIDIA)
- Ziyuan Cao - MS CS @ GT (January 2024 - May 2024, PhD student at OSU)
- Siddhartha Vemuri - MS CS @ GT (January 2023 - August 2023)
- Xianle Feng - MS CS @ GT (January 2024 - May 2024)
- Andrew Li - MS CS @ GT (January 2024 - May 2024)
- Arpan Parikh - MS CS @ GT (January 2024 - May 2024)
- Dhyan Gandhi - MS CS @ GT (January 2024 - May 2024, SWE at Amazon)
- Om Bhatt - MS CS @ GT (January 2024 - May 2024, Predoctoral Research @ UCI)
- Cheng Chang - MS CS @ Stanford (August 2023 - May 2024, now SWE at Google)
- Jason Liu - Undergraduate @ UPenn (August 2021 - August 2022)
- Damian Rene - Undergraduate @ Swarthmore (August 2021 - August 2022)

Spoken Languages

English (Native Proficiency), Hindi (Native Proficiency), Gujarati (Native Proficiency), Marathi (Expert Proficiency), and Sanskrit (Intermediate Proficiency).